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[23CSE305]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER-2024

FOUNDATIONS OF CLOUD COMPUTING

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5x2=10**		
1.	**a.**	Define cloud computing.	L1	CO1	**02**
	b.	Why are virtual machines important in cloud computing?	L1	CO2	**02**
	c.	Differentiate public and private cloud	L2	CO3	**02**
	d.	List out the pillars of cloud security	L2	CO4	**02**
	e.	State the applications of Hadoop in cloud computing	L1	CO5	**02**
		PART-B	**6X5=30**		
2.	**a.**	Summarize the Evolution of cloud computing	L2	CO1	**05**
		OR			
	b.	Explain in detail about parallel computing and list out its applications	L2	CO1	**05**
3.	**a.**	Identify the types of virtualization in cloud computing	L3	CO2	**05**
		OR			
	b.	Describe the REpresentational State Transfer (REST) system architecture	L3	CO2	**05**
4.	**a.**	Classify the types of cloud computing with example	L4	CO3	**05**
		OR			
	b.	Distinguish software as a service (SaaS) and platform as a service (PaaS)	L4	CO3	**05**
5.	**a.**	Why is SaaS Security important in cloud computing? Explain with necessary real-time example	L1	CO4	**05**
		OR			
	b.	List the Topologies used in the inter cloud management system with a neat diagram	L1	CO4	**05**
6.	**a.**	Draw the architecture of hadoop distributed file system (HDFS) and explain the functionalities of each node in a hadoop system	L3	CO5	**05**
		OR			
	b.	With neat sketch draw the architecture of Mapreduce and explain how mapper class is differ from the reduce class	L3	CO5	**05**

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7.	**a.**	Explain in detail about application development process on google app engine(GAE).	L2	CO5	**05**
		OR			
	b.	List out the key characteristics of cloud software environments.	L2	CO5	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Illustrate the characteristics of cloud computing with suitable example.	L2	CO1	**10**
		OR			
	b.	Demonstrate the cloud provisioning for allocation of resources and services from a cloud provider to a customer.	L2	CO1	**10**
9.	**a.**	Identify the components of service oriented architecture (SOA) and list its advantages.	L3	CO2	**10**
		OR			
	b.	Organize the importance of CPU virtualization and explain its types.	L3	CO2	**10**
10.	**a.**	Examine in detail about NIST Cloud Computing Reference Architecture model.	L4	CO3	**10**
		OR			
	b.	Classify the features and advantages of Amazon simple storage service (S3).	L4	CO3	**10**
11.	**a.**	Organize the security challenges faced by the cloud service provider in a deployment environment.	L3	CO4	**10**
		OR			
	b.	Consider you as a cloud service provider, write your opinion for the implementation and need of security governance for your organization.	L3	CO4	**10**
12.	**a.**	Discover the common components, advantages and disadvantages of google app engine (GAE).	L4	CO5	**10**
		OR			
	b.	Inspect the operation modes of Eucalyptus powerful open-source tool to build a private cloud in the organization.	L4	CO5	**10**
13.	**a.**	Explain in detail about the open stack independent components, which can be integrated to meet some specific needs.	L2	CO5	**10**
		OR			
	b.	Summarize the advantages of using cloud storage in IT industry.	L2	CO3	**10**